Eva Tan | +65 97796888 | etann.se@gmail.com | www.linkedin.com/in/evatann | https://evatann.github.io/portfolio/

Innovator and problem-solver eager to develop healthcare technologies with global impact.

EDUCATION

National University of Singapore, Singapore *Aug 2024 – Present*

Bachelor of Engineering (Biomedical Engineering) | Minor: Statistics

Temasek Polytechnic, Singapore *Apr 2021 – May 2024*

Diploma in Biomedical Engineering

SUMMARY OF SKILLS

- Programming & Software Development: Python, Java, JavaScript, C; Spyder, VS Code; Frontend/Backend Integration; Debugging.
- AI & Computational Tools: Machine Learning, OpenCV, AI Prototyping, Data Visualisation, KNIME, Tableau.
- Engineering & Technical Tools: Robotics Development, Circuit Analysis, Soldering, SOLIDWORKS, Autodesk EAGLE, Biomaterials, Medical Device Principles, GMP/GCP, PMS, CAPA, Design Control, Equipment Maintenance, Validation, Non-Conformance Handling, Technical Documentation.
- Laboratory & Experimental Techniques: Lab Safety, Titration, Basic Biochemistry/Chemistry Procedures, Instrument Handling.
- Research, Innovation & Project Management: R&D, Clinical Observation, Needs Assessment, Ideation, Prototyping, Product Design, Report Writing, Documentation.
- Professional & Interpersonal Skills: Communication, Collaboration, Leadership, Adaptability, Attention to Detail, Time Management, Stakeholder Engagement, Entrepreneurship, and Project management.
- Productivity & Office Tools: Microsoft Office 365.

WORK EXPERIENCE

Manufacturing Smart Lean Intern (Incoming) - WS Audiology, Singapore *May 2026 – Jul 2026*

- Selected for a Smart Lean Manufacturing internship focusing on process optimisation and operational efficiency in medical device production.
- Will support manufacturing improvement initiatives using Lean methodologies and data-driven analysis.

Research and Development Intern - Kumamoto National College of Technology, Japan *Apr 2023 – Aug 2023*

- Designed assistive technologies, including a sign-language gesture recognition system and head-controlled communication interface.
- Programmed in Python3 (OpenCV, MediaPipe, CVZone, NumPy, PyAutoGUI, PynPut, Math, and Pygame).
- Conducted prototyping, testing, and performance evaluations to improve communication tools for users with disabilities.
- Analysed technological trends and created conceptual designs to support innovation in assistive technology.
- Presented design concepts and incorporated feedback from Japanese supervisor across 4 review cycles, refining final solution to meet real-world usability standards.
- Gained international exposure and strengthened self-directed research, innovation, and cross-cultural collaboration skills.

Freelance Artist (Self-Employed) - Remote / Independent *Nov 2020 – Feb 2023*

- Created and sold original commissioned artworks (portraits, pet illustrations) priced at S$50–200 per piece, managing client briefs, revisions, and delivery independently.
- Built a small client base through self-directed marketing, handling all aspects of a creative business including pricing, communication, and fulfilment.

PROJECTS

Clinical Immersion & HealthTech Innovation – National University of Singapore *Jan 2026 – Ongoing*

- Conducting clinical immersions with a urology clinician at NUH; identified 8 unmet patient needs and focused on 1 for solution development.
- Applying the Stanford Biodesign framework to develop a medical technology solution, to be pitched to an audience of 14 clinicians and 28 students.

Personal WordPress Website – Self-Initiated *Dec 2025 – Ongoing*

- Designed a WordPress website to explore emerging medical technologies and devices.
- Research and analyse medical devices, healthcare systems, and innovation trends.
- Translate complex biomedical concepts into accessible content for non-technical readers.

Laser-Cut Microfluidic Device Design & Fabrication – National University of Singapore *Sep 2025 – Oct 2025*

- Prototyped and fabricated a three-layer microfluidic device using SolidWorks and laser cutting.
- Applied full CAD workflow: sketching, feature planning (Glass Box Method), advanced modelling (revolve, shell, mirror), and fully constrained assembly design with mates.
- Executed flow tests to evaluate channel integrity and mixing performance, informing design improvements.

Orbital NUS – National University of Singapore *Mar 2025 – Aug 2025*

- Led frontend development for AI-assisted cycle tracking app (React Native, Expo, Flask, Firebase, Google Gemini API).
- Integrated 3 AI-powered features (symptom logging, cycle predictions, chatbot); app achieved Artemis-level.

cAIreSync: Healthcare Monitoring System Concept – National University of Singapore *Feb 2025 – Mar 2025*

- Proposed an AI-powered CCTV monitoring system to detect falls, prolonged immobility, and unusual movements in elderly individuals.
- Designed a real-time alert pipeline with encrypted monitoring, scalable across many deployment environments (home, ward, care facility).

Innovative Solutions for ICDs – **Temasek Polytechnic, Singapore** *Oct 2023 – Feb 2024*

- Researched ICD complication types and co-developed patient-centred design improvements in a cross-disciplinary team of 4 (Biomedical + Mechatronics).

Data Analytics and Visualisation – Temasek Polytechnic, Singapore *Nov 2023 – Feb 2024*

- Built ML models in KNIME to predict loan approval likelihood, increasing accuracy of risk-based decision, for bank loan approval.
- Engineered features from a 851-row dataset and visualised key risk insights in Tableau, improving model interpretability.

WorldSkills Mobile Robotics – Temasek Polytechnic, Singapore *Oct 2021 – Feb 2023*

- Selected as 1 of 6 trainees; contributed to building an autonomous system for object detection and classification across 3 classes in Phase 1.
- Trained and optimized YOLO models and applied sensor filtering to enhance detection robustness.
- Programmed and tested autonomous navigation across 5+ competition-format environments, advancing to Phase 1 qualifier.

Acoustic Stethoscope Signal Processing System – Temasek Polytechnic, Singapore *May 2022 – Sep 2022*

- Designed and built a 4-stage biomedical signal processing circuit (pre-amp, feedback amplifier, Butterworth low-pass filter, audio amplifier) to amplify and filter heart/lung sounds from an electret microphone.
- Validated circuit performance across 10Hz–4kHz frequency sweep, achieving 23dB passband gain and ~996Hz low-pass cutoff, confirmed via oscilloscope and function generator testing.

CERTIFICATIONS AND LEARNING

Stanford University (Online)

AI in Healthcare Specialisation	*Ongoing*
Writing in Sciences	*Jan 2026*

AAPC

Coding and Medical Necessity	*Dec 2025*

Google

Foundations of User Experience (UX) Design	*Dec 2025*

Temasek Polytechnic

Diploma in Biomedical Engineering	*May 2024*
Certificate in University Preparatory Programme Mathematics	*Nov 2023*

Google

Google Analytics for Beginners	*Apr 2022*

AWARDS AND ACHIEVEMENTS

National University of Singapore

Certification of Outstanding Achievement (Project Management and Finance)	*Nov 2025*
NUS CSC Colours Award (Merit)	*Sep 2025*

Temasek Polytechnic

Edusave Skills Award	*Dec 2024*
ENGenius Award	*Feb 2024*
Director's List Award 2023	*Jun 2023*
TP ENG Peer Tutoring – Bronze Award	*Mar 2023*
WorldSkills Singapore 2023 – Mobile Robotics (Phase 1 Qualifier)	*Feb 2023*
Director's List Award 2022	*Jun 2022*
Edusave Certificate of Academic Achievement	*Jan 2022*

REFEREE CONTACTS

Available upon request